

1 Principles of Monte Carlo

1.1 Introduction Monte Carlo integration

Consider estimating $\alpha = \int_0^1 f(x) \, dx$. If $U \sim \text{Unif}(0, 1)$, then $\alpha = \mathbb{E}[f(U)]$. Given independent samples $U_1, \dots, U_n \sim \text{Unif}(0, 1)$, the Monte Carlo estimator is

$\hat{\alpha}_n = \frac{1}{n} \sum_{i=1}^n f(U_i)$.

By the strong law of large numbers, $\hat{\alpha}_n \rightarrow \alpha$ a.s. as $n \rightarrow \infty$. By the central limit theorem,

$\hat{\alpha}_n - \alpha \approx N(0, \sigma_f^2/n)$,

where $\sigma_f^2 = \text{var}(f(U))$. Thus the standard error is $\sigma_f/\sqrt{n}$. Halving the standard error requires increasing the number of samples by a factor of 4.

Dimensionality

For one-dimensional integration, Monte Carlo is usually not competitive with deterministic methods such as the trapezoidal rule, whose error is $O(n^{-2})$. The advantage of Monte Carlo is that its convergence rate is not directly sensitive to the dimension of the problem, while many deterministic integration methods suffer from the curse of dimensionality.

1.2 First Example: European Call Option

Risk-neutral valuation

Let $S(t)$ be the stock price at time t . A European call option with strike K and maturity T has payoff $(S(T) - K)^+ = \max\{0, S(T) - K\}$. Under the risk-neutral measure $\mathbb{Q}$, the time-0 price is

$V_0 = \mathbb{E}_0^{\mathbb{Q}}[e^{-rT} (S(T) - K)^+]$,

where r is the continuously compounded risk-free rate.

Stock-price dynamics

Under $\mathbb{Q}$, geometric Brownian motion satisfies $\frac{dS(t)}{S(t)} = r \, dt + \sigma \, dW(t)$, where $W(t)$ is a standard Brownian motion. The exact solution at time T is

$S(T) = S(0) \exp\left((r - \frac{1}{2}\sigma^2)T + \sigma W(T)\right)$.

Since $W(T) = \sqrt{T}Z$ with $Z \sim N(0, 1)$,

$S(T) = S(0) \exp\left((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z\right)$.

Simulation algorithm

For $i = 1, \dots, n$, generate $Z_i \sim N(0, 1)$, then set

$S_i(T) = S(0) \exp\left((r - \frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z_i\right)$,

$C_i = e^{-rT} (S_i(T) - K)^+$.

The Monte Carlo estimator is $\hat{C}_n = \frac{1}{n} \sum_{i=1}^n C_i$. It is unbiased, $\mathbb{E}[\hat{C}_n] = C$, and strongly consistent, $\hat{C}_n \rightarrow C$ with probability 1.

Confidence interval

The sample standard deviation of the simulated discounted payoffs is

$s_C = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (C_i - \hat{C}_n)^2}$.

An asymptotic $1 - \delta$ confidence interval is

$\hat{C}_n \pm z_{\delta/2} \frac{s_C}{\sqrt{n}}$.

Since s_C is estimated, one may replace the normal quantile with the corresponding t -quantile with $n - 1$ degrees of freedom.

Black-Scholes benchmark

The Black-Scholes value of a European call is

$BS(S, \sigma, T, r, K) = S\Phi(d_1) - e^{-rT}K\Phi(d_2)$,

where

$d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$,

$d_2 = \frac{\log(S/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$.

1.3 Simulation Layers

Layered construction

Monte Carlo derivative pricing is built in layers:

$U_1 \rightarrow Z_1 \rightarrow S_1(T)$ or $\{S_1(t_j)\}_{j=1}^n \rightarrow C_1 \rightarrow \hat{C}_n, s_C$.

The lower layers transform simple random variables into asset prices or paths. The upper layers transform paths into discounted payoffs and then into statistical summaries.

1.4 Second Example: Asian Option

Path dependence

An Asian option is path dependent. A discrete-average Asian call has payoff $(\bar{S} - K)^+$, where

$\bar{S} = \frac{1}{m} \sum_{i=1}^m S(t_j)$,

and $0 = t_0 < t_1 < \dots < t_m = T$. Its time-0 value is

$\mathbb{E}_0^{\mathbb{Q}}[e^{-rT} (\bar{S} - K)^+]$.

To estimate this value, simulate the path values $S(t_1), \dots, S(t_m)$, compute the average $\bar{S}$, discount the payoff, and average over independent replications.

1.5 Efficiency of Simulation Estimators

Central limit theorem

Suppose $\hat{C}_n = \frac{1}{n} \sum_{i=1}^n C_i$, where C_i are i.i.d., $\mathbb{E}_0^{\mathbb{Q}}[C_i] = C$, and $\text{var}(C_i) = \sigma_C^2 < \infty$. Then

$\frac{\hat{C}_n - C}{\sigma_C/\sqrt{n}} \Rightarrow N(0, 1)$,

or equivalently

$\sqrt{n}(\hat{C}_n - C) \Rightarrow N(0, \sigma_C^2)$.

Hence $\hat{C}_n - C \approx N(0, \sigma_C^2/n)$. For two unbiased estimators of the same quantity, the estimator with lower variance is preferred, other things being equal.

1.6 Bias

General principle

Variance reduction and faster computation are useful only if the estimator converges to the correct value. Some estimators are biased for every finite n , but become asymptotically unbiased as computational effort increases.

Ratio estimator

For i.i.d. samples (X_i, Y_i) , the estimator $\bar{X}/\bar{Y}$ for $\mathbb{E}[X]/\mathbb{E}[Y]$ is generally biased because

$\mathbb{E}\left[\frac{\bar{X}}{\bar{Y}}\right] \neq \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}$.

However,

$\frac{\bar{Y}}{\bar{X}} \rightarrow \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}$

with probability 1, and the normalized error is asymptotically normal. The bias becomes negligible as $n \rightarrow \infty$.

Model discretization error

If the exact geometric Brownian motion transition is replaced by the Euler recursion

$S((j+1)h) = S(jh) + rS(jh)h + \sigma S(jh)\sqrt{h}Z_{j+1}$,

then the simulated path distribution is not exactly the same as that implied by the continuous-time model. Option values estimated from this approximate dynamics are biased.

This model discretization bias can be reduced or eliminated by: (i) using the exact transition when available; (ii) decreasing the time step h , at the cost of more transitions per path.

Payoff discretization error

For a continuously averaged Asian option,

$\bar{S} = \frac{1}{T} \int_0^T S(u) \, du$.

Even exact simulation at a finite set of dates cannot compute the continuous average exactly. The payoff must be approximated, which creates payoff discretization error. This bias can be made small by using a smaller simulation time step, but computational cost increases.

Model discretization error comes from approximating the underlying dynamics. Payoff discretization error comes from approximating the payoff functional.

1.7 Mean Square Error

Bias-variance decomposition

For an estimator $\hat{\alpha}$ of α , the mean square error is

$\text{MSE}(\hat{\alpha}) = \mathbb{E}\left[(\hat{\alpha} - \alpha)^2\right]$.

It decomposes as

$\text{MSE}(\hat{\alpha}) = (\mathbb{E}[\hat{\alpha}] - \alpha)^2 + \mathbb{E}\left[(\hat{\alpha} - \mathbb{E}[\hat{\alpha}])^2\right]$,

or

$\text{MSE}(\hat{\alpha}) = \text{Bias}^2(\hat{\alpha}) + \text{Variance}(\hat{\alpha})$.

Thus MSE balances bias and variance.

2 Generating Random Numbers and Random Variables

2.1 Random Number Generation

Uniform inputs

Monte Carlo simulation is driven by apparently random numbers. Ideally we want $U_1, U_2, \dots$ such that each $U_i \sim \text{Unif}[0, 1]$ and the U_i are mutually independent. Independence is crucial: U_i should not be predictable from $U_1, \dots, U_{i-1}$.

A pseudorandom number generator produces a finite deterministic sequence $u_1, \dots, u_K \in [0, 1]$. A good generator makes short segments hard to distinguish from independent uniforms. Desirable properties: long period, reproducibility, speed, portability, and statistical randomness.

Linear congruential generator

A linear congruential generator is

$x_{i+1} = ax_i \bmod m$, $u_{i+1} = x_{i+1}/m$.

Here a is the multiplier, m the modulus, and $x_0 \in \{1, \dots, m - 1\}$ the seed. The modulo operation is $y \bmod m = y - \lfloor y/m \rfloor m$.

Once a value repeats, the whole sequence repeats. The generator has full period if it produces all $m - 1$ nonzero values before repeating. Large m alone is not enough: poor choices of a, m can produce short cycles.

2.2 General Sampling Methods

Uniform distribution

Assume independent uniforms satisfying $P(U_i \leq u) = 0$ if $u < 0$, $P(U_i \leq u) = u$ if $0 \leq u \leq 1$, and $P(U_i \leq u) = 1$ if $u > 1$. Sampling methods transform these uniforms into variables with desired distributions.

2.3 Inverse Transform Method

General method

To sample X with CDF F , set

$X = F^{-1}(U)$, $U \sim \text{Unif}[0, 1]$.

If F is not strictly increasing, use the generalized inverse

$F^{-1}(u) = \inf\{x : F(x) \geq u\}$.

This chooses the smallest compatible x . Validity follows from

$P(X \leq x) = P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$.

Flat sections of F have zero probability: if F is constant on $[a, b]$, then $P(a < X \leq b) = F(b) - F(a) = 0$.

Examples

Exponential with mean θ : $F(x) = 1 - e^{-x/\theta}$ for $x \geq 0$, so

$X = -\theta \log(1 - U)$, equivalently $X = -\theta \log U$.

Rayleigh example:

$F(x) = 1 - e^{-2x(x-b)}$, $x \geq b$,

so the inverse-transform sample is

$X = \frac{b}{2} + \frac{\sqrt{b^2 - 2\log U}}{2}$.

Discrete distribution: For outcomes $c_1 < \dots < c_n$ with probabilities p_i , define $q_i = 0$ and

$q_i = \sum_{j=1}^i p_j$.

Generate U , find K with $q_{K-1} < U \leq q_K$, and set $X = c_K$. Binary search can implement the lookup.

Conditional sampling

To sample X conditional on $a < X \leq b$, with $F(a) < F(b)$, generate

$V = F(a) + [F(b) - F(a)]U$.

Then $F^{-1}(V)$ has the conditional distribution because

$P(F^{-1}(V) \leq x) = \frac{F(x) - F(a)}{F(b) - F(a)}$.

2.4 Acceptance-Rejection Method

General method

To sample from a target density f on $\mathcal{X}$, choose a convenient proposal density g and a constant c such that

$f(x) \leq c g(x)$, $x \in \mathcal{X}$.

Generate $X \sim g$ and $U \sim \text{Unif}[0, 1]$. Accept X if

$U \leq \frac{f(X)}{c g(X)}$;

otherwise reject and repeat. If Y is the accepted value, then $P(Y \in A) = P(X \in A \mid U \leq f(X)/(c g(X)))$.

The acceptance probability is

$P(U \leq f(X)/(c g(X))) = \int_{\mathcal{X}} \frac{f(x)}{c g(x)} g(x) \, dx = 1/c$.

Hence

$P(Y \in A) = c \int_A \frac{f(x)}{c g(x)} g(x) \, dx = \int_A f(x) \, dx$.

Thus Y has density f . The number of proposals until acceptance is geometric with mean c , so smaller c and tighter bounds are better.

Beta distribution

The beta density on $[0, 1]$ is

$f(x) = \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} x^{\alpha_1-1} (1 - x)^{\alpha_2-1}$.

Here

$B(\alpha_1, \alpha_2) = \int_0^1 x^{\alpha_1-1} (1 - x)^{\alpha_2-1} \, dx = \frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1 + \alpha_2)}$,

and

$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} \, dx$, $\text{Re}(z) > 0$.

For integer $n > 0$, $\Gamma(n) = (n - 1)!$. If $\alpha_1, \alpha_2 \geq 1$ and at least one exceeds 1, the mode is $\frac{\alpha_1-1}{\alpha_1+\alpha_2-2}$.

Let c be the maximum value of f and use $g(x) = 1$ on $[0, 1]$. Generate U_1, U_2 until $cU_2 \leq f(U_1)$, then return U_1 .

Normal from double exponential

For the double exponential density $g(x) = e^{-|x|}/2$ and standard normal density $f(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$,

the ratio satisfies

$\frac{f(x)}{g(x)} = \sqrt{\frac{\pi}{2}} e^{-x^2/2+|x|} \leq \sqrt{\frac{\pi}{2}} \approx 1.3155 \approx c$.

The rejection test may be written as

$u > \exp\left(-\frac{1}{2}x^2 + |x| - \frac{1}{2}\right) = \exp\left(-\frac{1}{2}(|x| - 1)^2\right)$.

Using symmetry: generate U_1, U_2, U_3 , set $X = -\log U_1$, reject if $U_2 > \exp[-0.5(X - 1)^2]$, then set $X \leftarrow -X$ if $U_3 \leq 0.5$.

2.5 Normal Random Variables and Vectors

Multivariate normal

A d -dimensional normal is written $N(\mu, \Sigma)$, where μ is a d -vector and Σ is a $d \times d$ covariance matrix. A covariance matrix must be symmetric and positive semidefinite:

$x^\top \Sigma x \geq 0$, $x \in \mathbb{R}^d$.

If Σ is positive definite, the density is

$\phi_{\mu, \Sigma}(x) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right]$.

If $X \sim N(\mu, \Sigma)$, then $X_i \sim N(\mu_i, \sigma_i^2)$ with $\sigma_i^2 = \Sigma_{ii}$, and

$\text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] = \Sigma_{ij}$,

$\rho_{ij} = \frac{\Sigma_{ij}}{\sigma_i \sigma_j}$, equivalently $\Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$.

If Σ is positive semidefinite but singular, define $X = \mu + AZ$ with $Z \sim N(0, I_d)$ and $AA^\top = \Sigma$. Then X has no full d -dimensional density when $\text{rank}(\Sigma) < d$.

Useful properties

Linear transformation: if $X \sim N(\mu, \Sigma)$, then

$AX \sim N(A\mu, A\Sigma A^\top)$.

Conditioning formula: for a partitioned normal vector with $\Sigma_{[22]}$ full rank,

$(X_{[1]} \mid X_{[2]} = x) \sim N(\mu_{[1]} + \Sigma_{[12]}\Sigma_{[22]}^{-1}(x - \mu_{[2]}), \Sigma_{[11]} - \Sigma_{[12]}\Sigma_{[22]}^{-1}\Sigma_{[21])}$.

Moment generating function:

$\mathbb{E}[\exp(\theta^\top X)] = \exp(\mu^\top \theta + \frac{1}{2}\theta^\top \Sigma \theta)$.

2.6 Generating Normals

Box-Muller method

The Box-Muller method generates a bivariate standard normal. If $Z \sim N(0, I_2)$, then $R = Z_1^2 + Z_2^2$ is exponential with mean 2:

$P(R \leq s) = 1 - e^{-s/2}$.

Given $R, (Z_1, Z_2)$ is uniformly distributed on the circle of radius $\sqrt{R}$. Generate

independent U_1, U_2 , set

$R = -2\log U_1$, $V = 2\pi U_2$,

and return

$Z_1 = \sqrt{R} \cos V$, $Z_2 = \sqrt{R} \sin V$.

Multivariate normals

To sample $X \sim N(\mu, \Sigma)$, generate $Z \sim N(0, I_d)$, find A such that

$AA^\top = \Sigma$, and set $X = \mu + AZ$. Then $X \sim N(\mu, \Sigma)$.

2.7 Cholesky Factorization

Positive definite case

Cholesky represents $\Sigma = AA^\top$ with A lower triangular. For

$\Sigma = \begin{pmatrix} \sigma_1^2 & & \\ \sigma_1 \sigma_2 \rho & \sigma_2^2 \rho^2 \\ \sigma_1 \sigma_2 \rho & \sigma_2^2 \rho^2 \end{pmatrix}$,

a Cholesky factor is

$A = \begin{pmatrix} \sigma_1 & & 0 \\ \rho \sigma_2 & \sqrt{1 - \rho^2} \sigma_2 \end{pmatrix}$.

Thus

$X_1 = \mu_1 + \sigma_1 Z_1$,

$X_2 = \mu_2 + \sigma_2 \rho Z_1 + \sigma_2 \sqrt{1 - \rho^2} Z_2$.

For general d , use

$\Sigma_{ij} = \sum_{k=1}^i A_{ik} A_{jk}$, $j \leq i$.

The recursive formulas are

3.6 Multi-Asset GBM

Correlated assets

Let X_t be standard Brownian motions with correlations ρ_{ij} . A multidimensional GBM is

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i dX_i(t), \quad i = 1, \dots, d.$$

Define $\Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$. Then $(\sigma_1 X_1, \dots, \sigma_d X_d) \sim \text{BM}(0, \Sigma)$ and $S = (S_1, \dots, S_d) \sim \text{GBM}(\mu, \Sigma)$.

The price covariance is

$$\text{Cov}[S_i(t), S_j(t)] = S_i(0)S_j(0)e^{(\mu_i+\mu_j)t} \left(\rho_{ij} \sigma_i \sigma_j t - 1 \right).$$

If $AA^T = \Sigma$ and a_i is the i th row of A ,

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + a_i dW(t) = \mu_i dt + \sum_{j=1}^d A_{ij} dW_j(t).$$

Simulation and payoffs

With independent $Z_t = (Z_{k1}, \dots, Z_{kd}) \sim N(0, I)$,

$$S_i(t_{k+1}) = S_i(t_k) \exp \left(\left[\mu_i - \frac{1}{2} \sigma_i^2 \right] \Delta t_k + \sqrt{\Delta t_k} \sum_{j=1}^d A_{ij} Z_{kj+1, j} \right).$$

Under the risk-neutral measure, $\mu_i = r - \delta_i$. Typical multi-asset payoffs:

Spread option:

$$\mathbb{E}[|S_1(T) - S_2(T)| - K]^+.$$

Basket option:

$$(|c_1 S_1(T) + c_2 S_2(T) + \dots + c_d S_d(T) - K|)^+.$$

Outperformance option:

$$(\max(c_1 S_1(T), c_2 S_2(T), \dots, c_d S_d(T)) - K)^+.$$

Two-asset barrier example:

$$\mathbf{1}_{[\min_{i=1, \dots, n} S_2(t_i) < b]} (K - S_1(T))^+.$$

4 Variance Reduction Techniques

4.1 Control Variates

Basic method

Let $Y_1, \dots, Y_b$ be i.i.d. outputs for estimating $\mathbb{E}[Y]$, and suppose X_i is generated with Y_i and $\mathbb{E}[X]$ is known. For fixed b define

$$Y_i(b) = Y_i - b(X_i - \mathbb{E}[X]),$$

and estimate

$$\bar{Y}(b) = \bar{Y} - b(\bar{X} - \mathbb{E}[X]) = \frac{1}{b} \sum_{i=1}^b (Y_i - b(X_i - \mathbb{E}[X])).$$

This is unbiased for all b . Its per-replication variance is

$$\sigma^2(b) = \text{var}[Y_i(b)] = \sigma_Y^2 - 2b\sigma_X \rho_{XY} + b^2 \sigma_X^2.$$

The optimal coefficient is

$$b^* = \frac{\sigma_Y}{\sigma_X} \rho_{XY} = \frac{\text{Cov}(X, Y)}{\text{var}(X)}.$$

The resulting variance ratio relative to plain Monte Carlo is

$$\frac{\text{var}[\bar{Y} - b^*(\bar{X} - \mathbb{E}[X])]}{\text{var}[\bar{Y}]} = 1 - \rho_{XY}^2.$$

Thus speed-up is approximately $1/(1 - \rho_{XY}^2)$. The sign of ρ_{XY} is absorbed by b^* ; only $|\rho_{XY}|$ matters.

Estimated coefficient and CI

In practice use the sample regression coefficient

$$\hat{b}_n = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}.$$

Then $\hat{b}_n \rightarrow b^*$ a.s. and

$$\bar{Y}(\hat{b}_n) = \frac{1}{b} \sum_{i=1}^n \{Y_i - \hat{b}_n(X_i - \mathbb{E}[X])\}.$$

Using $\hat{b}_n$ creates finite-sample bias, but has the same asymptotic precision as using b^* . For fixed b ,

$$\frac{\bar{Y}(b) - \mathbb{E}[\bar{Y}]}{\sigma(b)/\sqrt{m}} \Rightarrow N(0, 1),$$

with sample standard deviation

$$s(b) = \left[\frac{1}{n-1} \sum_{i=1}^n (Y_i(b) - \bar{Y}(b))^2 \right]^{1/2}.$$

Hence an asymptotic CI is $\bar{Y}(b) \pm z_{\delta/2} s(b)/\sqrt{n}$. Also

$$\frac{\bar{Y}(\hat{b}_n) - \mathbb{E}[\bar{Y}]}{s(\hat{b}_n)/\sqrt{n}} \Rightarrow N(0, 1).$$

Financial controls

Any martingale with known initial value is a natural control. Under risk-neutral pricing with constant r , $e^{-rT} S(t)$ is a martingale, so $\mathbb{E}[e^{-rT} S(T)] = S(0)$. For discounted payoff Y ,

$$\frac{1}{b} \sum_{i=1}^n \{Y_i - b[e^{-rT} S_i(T) - S(0)]\}$$

uses the stock as control. For call, this works best near $K = 0$ and may weaken for deep OTM strikes. A stronger control can be a tractable option. For Asian options,

$$\bar{S}_A = \frac{1}{n} \sum_{i=1}^n S(t_i), \quad \bar{S}_G = \left(\prod_{i=1}^n S(t_i) \right)^{1/n}.$$

Under GBM, $\bar{S}_G$ is lognormal and the geometric-average Asian option can be priced analytically; its payoff is often highly correlated with the arithmetic-average Asian payoff.

4.2 Antithetic Variates

Estimator

Antithetic variates pair inputs to create negative dependence while preserving the marginal law. If $U \sim \text{Unif}[0, 1]$, use U and $1 - U$; for a symmetric normal input, use Z and $-Z$.

Let $(Y_i, \bar{Y}_i)$ be i.i.d. pairs with the same marginal distribution. The estimator is

$$Y_{\mathcal{A}\mathcal{V}} = \frac{1}{n} \sum_{i=1}^n Y_i + \sum_{i=1}^n \bar{Y}_i = \frac{1}{n} \sum_{i=1}^n \frac{Y_i + \bar{Y}_i}{2}.$$

Apply the CLT to pair averages:

$$\frac{Y_{\mathcal{A}\mathcal{V}} - \mathbb{E}[Y]}{\sigma_{\mathcal{A}\mathcal{V}}/\sqrt{n}} \Rightarrow N(0, 1), \quad \sigma_{\mathcal{A}\mathcal{V}}^2 = \text{var} \left(\frac{Y_1 + \bar{Y}_1}{2} \right).$$

A CI is $Y_{\mathcal{A}\mathcal{V}} \pm z_{\delta/2} s_{\mathcal{A}\mathcal{V}}/\sqrt{n}$, where $s_{\mathcal{A}\mathcal{V}}$ is the sample sd of pair averages.

When it helps

Compare n antithetic pairs with $2n$ independent samples. Since

$$\text{var}[Y_1 + \bar{Y}_1] = 2 \text{var}[Y_1] + 2 \text{Cov}(Y_1, \bar{Y}_1),$$

antithetic sampling reduces variance iff $\text{Cov}(Y_1, \bar{Y}_1) < 0$. Monotone outputs often satisfy this when paired by $U, 1 - U$ or $Z, -Z$.

GBM call example

For a call, pair Z_t and $-Z_t$:

$$S_t^+ = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z_t \right),$$

$$S_t^- = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) T - \sigma \sqrt{T} Z_t \right).$$

Set $X_t = e^{-rT} (S_t^+ - K)^+$ and $Y_t = e^{-rT} (S_t^- - K)^+$. Then

$$\hat{v} = \frac{1}{n} \sum_{i=1}^n \frac{X_i + Y_i}{2},$$

$$SE = \left[\frac{1}{n(n-1)} \prod_{i=1}^n \left(\frac{X_i + Y_i}{2} \right)^2 - n \hat{v}^2 \right]^{1/2}.$$

4.3 Stratified Sampling

Basic method

Let $A_1, \dots, A_K$ be disjoint strata with probabilities p_i . Then

$$\mathbb{E}[Y] = \sum_{i=1}^K p_i \mathbb{E}[Y \mid Y \in A_i].$$

With proportional allocation $n_i = np_i$ and conditional draws $Y_{ij} \sim (Y \mid Y \in A_i)$,

$$\hat{Y} = \sum_{i=1}^K p_i \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij} = \frac{1}{n} \sum_{i=1}^K \sum_{j=1}^{n_i} Y_{ij}.$$

Stratification removes the random variation in the number of observations landing in each stratum.

General allocation

Often strata are defined by another variable X . If A_i partition the support of X ,

$$\mathbb{E}[Y] = \sum_{i=1}^K p_i \mathbb{E}[Y \mid X \in A_i].$$

For $n_i \geq 1$, $\sum_i n_i = n$, and $q_i = n_i/n$,

$$\hat{Y} = \sum_{i=1}^K p_i \frac{n_i}{\hat{n}_i} \sum_{j=1}^{n_i} Y_{ij} = \frac{1}{n} \sum_{i=1}^K \frac{p_i}{q_i} \sum_{j=1}^{n_i} Y_{ij}.$$

Let $\mu_i = \mathbb{E}[Y \mid X \in A_i]$ and $\sigma_i^2 = \text{var}(Y \mid X \in A_i)$. Then $\hat{Y}$ is unbiased and

$$\text{var}[\hat{Y}] = \sum_{i=1}^K p_i \frac{\sigma_i^2}{q_i} \frac{p_i^2}{\hat{n}_i} = \frac{\sigma^2(q)}{n}, \quad \sigma^2(q) = \sum_{i=1}^K p_i^2 \frac{\sigma_i^2}{q_i}.$$

With fixed K , $\sqrt{n}(\hat{Y} - \mu) \Rightarrow N(0, \sigma^2(q))$, and

$$s^2(q) = \sum_{i=1}^K p_i^2 \frac{\sigma_i^2}{q_i} \hat{\sigma}_i^2$$

estimates $\sigma^2(q)$.

Uniform and inverse-transform strata

For $U \sim \text{Unif}[0, 1]$, equal strata are $A_i = ((i-1)/n, i/n)$. With $U_i \sim \text{Unif}[0, 1]$,

$$V_i = \frac{i-1}{n} + \frac{U_i}{n}, \quad Y = \frac{1}{n} \sum_{i=1}^n f(V_i).$$

For a distribution with CDF F , choose probabilities p_i and quantiles $a_i = F^{-1}(p_1 + \dots + p_i)$. Conditional simulation in stratum $(a_{i-1}, a_i]$ uses

$$V = F(a_{i-1}) + U \{F(a_i) - F(a_{i-1})\}, \quad F^{-1}(V) \sim (Y \mid Y \in A_i).$$

Proportional and optimal allocation

For proportional allocation $q_i = p_i$,

$$\sigma^2(q) = \sum_{i=1}^K p_i \sigma_i^2.$$

Plain variance decomposes as

$$\text{var}(Y) = \sum_{i=1}^K p_i \sigma_i^2 + \sum_{i=1}^K p_i \mu_i^2 - \left(\sum_{i=1}^K p_i \mu_i \right)^2.$$

By Jensen, proportional stratification cannot increase variance, with strict gain unless all μ_i are equal. Optimizing $\sigma^2(q)$ gives

$$q_i^* = \frac{p_i \sigma_i}{\sum_{i=1}^K p_i \sigma_i}, \quad \sigma^2(q^*) = \left(\sum_{i=1}^K p_i \sigma_i \right)^2.$$

Thus allocate samples proportional to stratum probability times within-stratum sd.

Call option strata

For a GBM call, write $W_t = \sqrt{T} \Phi^{-1}(U)$ with $U \sim \text{Unif}[0, 1]$ and

$$h(u) = \left(S_0 \exp \left\{ -\frac{1}{2} \sigma^2 T + \sigma \sqrt{T} \Phi^{-1}(u) \right\} - e^{-rT} K \right)^+.$$

With k equal strata, $U_{ij} = (i-1)/k + V_{ij}/k$, $\hat{\mu}_i = n_i^{-1} \sum_{j=1}^{n_i} h(U_{ij})$, and

$$\hat{\mu} = \frac{1}{k} \sum_{i=1}^k \hat{\mu}_i, \quad SE = \frac{1}{k} \left(\sum_{i=1}^k \frac{\hat{\sigma}_i^2}{n_i} \right)^{1/2}.$$

4.4 Importance Sampling

Change of measure

To estimate $\mu = \mathbb{E}[h(X)]$ with density f , choose an alternative density g such that $g(x) = 0$ implies $f(x) = 0$. If $Y \sim g$,

$$\mu = \int h(x) f(x) dx = \int h(x) \frac{f(x)}{g(x)} g(x) dx = \mathbb{E}_g \left[h(Y) \frac{f(Y)}{g(Y)} \right].$$

The likelihood ratio is $f(Y)/g(Y)$, and the estimator averages $h(Y_i) f(Y_i)/g(Y_i)$. In a discrete model, use $h(Y_i) p(Y_i)/\bar{p}(Y_i)$.

Variance and zero-variance density

The IS estimator has variance

$$\frac{1}{n} \text{var}_g \left[h(Y) \frac{f(Y)}{g(Y)} \right].$$

For $h \geq 0$, the zero-variance density is

$$g^*(x) = c h(x) f(x), \quad \frac{1}{c} = \int h(x) f(x) dx = \mu.$$

It is unusable directly because it requires μ , but it says g should mimic $h(x) f(x)$.

Normal mean shift and options

For $X \sim N(0, 1)$, choose $Y \sim N(\theta, 1)$. The likelihood ratio is

$$\frac{f(x)}{g(x)} = \exp \left\{ -\theta x + \frac{1}{2} \theta^2 \right\}.$$

Mode matching sets $\theta = \arg \max_x h(x) f(x)$. For a binary call,

$$\mathbb{E}[e^{-rT} \mathbf{1}_{\{S_T \geq K\}}] = \mathbb{E}[e^{-rT} \mathbf{1}_{\{X \geq b\}}],$$

where

$$b = \frac{\log(K/S_0)}{\sigma \sqrt{T}} - \frac{(r - \sigma^2/2)T}{\sigma \sqrt{T}}.$$

Here $h(x) = e^{-rT} \mathbf{1}_{\{x \geq b\}}$, $x^* = \max\{b, 0\}$, and with $Y \sim N(x^*, 1)$ the output is

$$h(Y) \frac{f(Y)}{g(Y)} = \begin{cases} 0, & Y < b, \\ \exp\{-rT - x^* Y + (x^*)^2/2\}, & Y \geq b. \end{cases}$$

For a vanilla call,

$$h(x) = \left(S_0 \exp \left\{ -\frac{1}{2} \sigma^2 T + \sigma \sqrt{T} x \right\} - e^{-rT} K \right)^+,$$

and the shift pushes sampling toward exercise states.

Exponential tilting

For density f , the exponential tilt family is

$$f_\theta(x) = \frac{e^{\theta x} f(x)}{\int e^{\theta x} f(x)}, \quad \theta \in \mathbb{R}.$$

For a pmf p ,

$$p_\theta(x) = \frac{e^{\theta x} p(x)}{\sum_x e^{\theta x} p(x)}.$$

For normal X , this is mean shifting. For rare large losses, take $\theta > 0$.

Bernoulli rare event

Let independent defaults satisfy $P(X_k = 1) = p_k$ and $L = \sum_{k=1}^m X_k$. To estimate $P(L > x)$ with $x > \mathbb{E}[L]$, tilt each Bernoulli:

$$\bar{p}_k = P(Y_k = 1) = \frac{p_k e^\theta}{1 + p_k(e^\theta - 1)}, \quad \bar{L} = \sum_{k=1}^m Y_k.$$

The IS output is

$$H = \mathbf{1}_{\{\bar{L} > x\}} \prod_{k=1}^m \left(\frac{p_k}{\bar{p}_k} \right)^{Y_k} \left(\frac{1 - p_k}{1 - \bar{p}_k} \right)^{1 - Y_k}.$$

With

$$\phi(\theta) = \sum_{k=1}^m \log[1 + p_k(e^\theta - 1)],$$

one has $\mathbb{E}[H^2] \leq e^{-2\phi(\theta) + 2\phi(0)}$. Choose $\theta > 0$ by solving $\phi'(\theta) = x$.

4.5 Cross-Entropy Method

Principle

Cross-entropy selects a parametric IS density f_θ close to the zero-variance density

$$g^*(x) = \frac{1}{b} h(x) f(x).$$

Minimize

$$R(g^* \| f_\theta) = \int \log \left(\frac{g^*(x)}{f_\theta(x)} \right) g^*(x) dx,$$

equivalently maximize

$$\int h(x) f(x) \log f_\theta(x) dx.$$

The population first-order equation is

$$\mathbb{E} \left[h(X) \frac{\partial}{\partial \theta} \log f_\theta(X) \right] = 0.$$

Basic algorithm

Using pilot samples $X_1, \dots, X_N \sim f$, solve

$$\frac{1}{N} \sum_{k=1}^N h(X_k) \frac{\partial}{\partial \theta} \log f_\theta(X_k) = 0.$$

For $f = N(0, I_m)$ and $f_\theta = N(\theta, I_m)$,

$$\hat{\theta} = \frac{\sum_{k=1}^N h(X_k) X_k}{\sum_{k=1}^N h(X_k)}.$$

Then sample $Y_i \sim f_{\hat{\theta}}$ and average

$$H_i = h(Y_i) \frac{f(Y_i)}{f_{\hat{\theta}}(Y_i)}.$$

For a one-dimensional normal shift, $H_i = h(Y_i) \exp\{-\hat{\theta} Y_i + \hat{\theta}^2/2\}$. For very rare events, the basic method may fail because $\sum_k h(X_k) = 0$ in the pilot run.

Iterative cross-entropy

Let $t_\theta(x) = f(x)/f_\theta(x)$. If pilot samples are drawn from f_v , then maximize $\mathbb{E}_v[h(Y) t_v(Y) \log f_\theta(Y)]$, and solve

$$\mathbb{E}_v \left[h(Y) t_v(Y) \frac{\partial}{\partial \theta} \log f_\theta(Y) \right] = 0.$$

At iteration j , sample $Y_k \sim f_{\hat{\theta}_j}$ and update from

$$\frac{1}{N} \sum_{k=1}^N h(Y_k) t_{\hat{\theta}_j}(Y_k) \frac{\partial}{\partial \theta} \log f_{\hat{\theta}_j}(Y_k) = 0.$$

For $N(\theta, I_m)$,

$$\hat{\theta}_{j+1} = \frac{\sum_{k=1}^N h(Y_k) t_{\hat{\theta}_j}(Y_k) Y_k}{\sum_{k=1}^N h(Y_k) t_{\hat{\theta}_j}(Y_k)}.$$

Here

$$t_{\hat{\theta}_j}(Y_k) = \exp(-(\hat{\theta}_j, Y_k) + \|\hat{\theta}_j\|^2/2).$$

Usually only a few iterations are needed; convergence means the updates oscillate in a small range rather than become deterministic.

5 Discretization Methods

5.1 Euler And Milstein Schemes

Euler scheme

For the SDE $dX(t) = a(X(t)) dt + b(X(t)) dW(t)$, with $X \in \mathbb{R}^d$ and $W \in \mathbb{R}^m$, the Euler approximation on $t_i = ih$ is

$$\hat{X}(i+1) = \hat{X}(i) + a(\hat{X}(i))h + b(\hat{X}(i))\Delta W(i)/\sqrt{h} Z_{i+1},$$

where Z_i are independent m -dimensional standard normals. Euler freezes the drift and diffusion integrands on each interval. The drift is order h , while the diffusion increment is order $\sqrt{h}$, so a first refinement focuses on the diffusion term.

Scalar Milstein refinement

In the scalar case, Ito's formula gives

$$db(X(t)) = [b'(X(t))a(X(t)) + \frac{1}{2} b''(X(t))b^2(X(t))] dt$$

$$+ b'(X(t))b(X(t))dW(t).$$

Dropping the higher-order drift part,

$$b(X(t)) \approx b(X(t)) + b'(X(t))b(X(t))[W(t) - W(t)].$$

With $\Delta W = W(t+h) - W(t)$,

$$f_t^{t+h} b$$

Multivariate normal score

If

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6.4 Gamma And Examples

LRM gamma

If

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 is the second density derivative, then

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estimates

d

2

E
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f
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d

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2

,

 under valid derivative-expectation interchanges.

For Black-Scholes gamma,

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Replace

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 when using the usual GBM simulation. Higher derivati- ves introduce higher powers and often larger variance. For

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X
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4

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Artificial-dynamics control for delta

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Then call delta is

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Use Euler for

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 and an artificial constant-volatility process with the same

Z

i

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Y

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2

)
Δ

t

i

+
σ

√

Δ

t

i

Z

i
+
1

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The control is

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e

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y
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Use output

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b
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V
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]
,

 typically with

σ
=
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.

Euler sensitivity recursion

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 and

V
(

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)
=
E
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−
r
T

h
(

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]
,

Δ
=
E
⁡
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h
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d

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S

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Euler simulation:

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S

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S

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+
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=

S

i

+

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S

i

Δ

t

i

+
σ
(

S

i

)

√

Δ

t

i

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i
+
1

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Define

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i

=
∂

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i

/
∂

S

0

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 Then

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0

=
1,
    

Δ

i
+
1

=

Δ

i

+

r

Δ

i

Δ

t

i

+
σ
′
(

S

i

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Δ

i

√

Δ

t

i

Z

i
+
1

.

Estimate delta by averaging

e

−
r
T

h
′
(

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T

)

Δ

i

T

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7 Pricing American Options

7.1 Backward Induction And LSM

American exercise problem

Simulation generates paths forward, but an American option needs a backward exercise rule. For a put with exercise dates

t

1

<
⋯
<

t

n

=
T

 and

t

j
+
1

−

t

j

=
Δ
t
,

 simulate paths

{

S

(

t

1

)
,
.
.
.
,
S
(

t

n

)
}

 and define

Y

j

 as the cash flow from

t

j

 onward under the selected policy.

At maturity,

Y

n

=
max
⁡
(
K
−
S
(

t

n

)
,
0
)
.

For earlier dates, the generic cash-flow recursion is

Y

j

=

{

K
−
S
(

t

j

)
,
    
exercise,

e

−
r
Δ

t

Y

j
+
1

,
    
continue.

There is no exercise right before

t

1

,

 so the American put is priced as a claim paying

Y

1

 at

t

1

:

P

A

(
0
,
S
)
=
E
⁡
[

e

−
r

t

1

Y

1

|

S

0

=
S
]
,
    
P

A

(
0
,
S
)
=

1
N

∑

N

i
=
1

e

−
r

t

i

Y

1

i

(

t

i

)
.

Continuation value

At

t

j

,

 the decision must use information available at

t

j

.

 Compare immediate exercise

K
−
S
(

t

j

)

 with the conditional discounted continuation value

f

j

(
S
(

t

j

)
)
=
E
ˆ
[

e

−
r
Δ

t

Y

j
+
1

|

S
(

t

j

)
]
.

Thus

Y

j

=

{

K
−
S
(

t

j

)
,
    
K
−
S
(

t

j

)
≥

f

j

(
S
(

t

j

)
)
,

e

−
r
Δ

t

Y

j
+
1

,
    
K
−
S
(

t

j

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<

f

j

(
S
(

t

j

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)
.

For out-of-the-money put paths,

K
−
S
(

t

j

)
<
0
,

 set

f

j

(
S
(

t

j

)
)
=
0

 and continue:

Y

j

=

e

−
r
Δ

t

Y

j
+
1

.

Least-squares approximation

Longstaff-Schwartz estimates

f

j

 from simulated paths by least squares. Use only in-the-money paths in the regression, because only they are close to the exercise decision. Approximate

f

j

(
S
(

t

j

)
)
≈

∑

N

k
=
0

a

j
k

S
(

t

j

)

k

.

Regress the realized discounted continuation value

e

−
r
Δ

t

Y

j
+
1

 on basis functions of

S
(

t

j

)
.

 Degree-3 polynomials are often reasonable; Hermite, Laguerre, and Legendre bases are common alternatives, and prices are usually fairly robust to the basis choice.

7.2 Three-Date American Put Example

Setup

Let

S
(
t
)

 follow risk-neutral GBM with

S
(
0
)
=
10
,
r
=
3
%
,

 dividend yield

δ
=
0
,

σ
=
40
%
,

 strike

K
=
12
,

 maturity

T
=
1
,

 and exercise dates

1

/

3
,
2

/

3
,
1
.

 Simulate with

S
(
t
+
Δ
t
)
=
S
(
t
)
exp
⁡
[
(
r
−

σ

2

)
Δ
t
+
σ
d

W

(

t

)
]
.

Using the 8 sample paths in the lecture, the terminal payoff

Y

3

=
max
⁡
(
K
−
S
(
1
)
,
0
)

 gives the European put estimate 2.4343, a lower bound for the Ameri- can put.

**Step at

t
=
2

/

3**

For in-the-money paths, regress

Y

3

e

−
r
Δ
t

 on

S
(
2

/

3
)

 using a quadratic:

Y

3

e

−
r
Δ
t

=

a

0

+

a

1

S
(
2

/

3
)
+

a

2

[
S
(
2

/

3
)
]

2

+
ε.

The fitted continuation value is

f

2

(
S
)
=
−
82.5347
+
17.7788
S
(
2

/

3
)
−
0.9063
[
S
(
2

/

3
)

]

2

.

Then

Y

2

=

{

K
−
S
(
2

/

3
)
,
  
K
−
S
(
2

/

3
)
≥

f

2

(
S
(
2

/

3
)
)
,

e

−
r
Δ

t

Y

3

,
    
otherwise.

Out-of-the-money paths continue, so

Y

2

=

e

−
r
Δ

t

Y

3

 for those paths.

**Step at

t
=
1

/

3**

Repeat the regression using

Y

2

e

−
r
Δ
t

 and

S
(
1

/

3
)
.

 The fitted continuation value is

f

1

(
S
)
=
−
8.9488
+
3.3104
S
(
1

/

3
)
−
0.2036
[
S
(
1

/

3
)

]

2

.

Use

Y

1

=

{

K
−
S
(
1

/

3
)
,
  
K
−
S
(
1

/

3
)
≥

f

1

(
S
(
1

/

3
)
)
,

e

−
r
Δ

t

Y

2

,
    
otherwise.

Finally estimate the current price by averaging

e

−
r
Δ
t

Y

1

. In the lecture example,

P

A

=
3.0919
,

 which is above the European estimate 2.4343.

8 Efficient MC For Value-At-Risk

8.1 VaR And Delta-Gamma

Loss and VaR

Let

S

t

 be market prices/rates,

Δ
t

 the risk horizon,

Δ
S

 the risk-factor move, and

S
(
t
,
r
)

 the portfolio value. The horizon loss is

L
=
−
Δ
V
=
V
(
s
,
t
)
−
V
(
s
+
Δ
S
,
t
+
Δ
t
)
,
    

P

L

(
x
)
=
P
(
L
<
x
)
.

For tail probability

p
,

 VaR

x

p

 satisfies

1
−

F

L

(

x

p

)
=
P
(
L
>

x

p

)
=
p
.

For 99% VaR,

p
=
0.01
.

 VaR is not coherent because subadditivity can fail.

RiskMetrics

Assume

Δ
S
∼
N
(
0
,

Σ

S

)

 and linear portfolio response

Δ
V
=
δ

⊤

Δ
S
.

 Then

L
∼
N
(
0
,

σ

L

2

)
,
    

σ

L

2

=
δ

⊤

Σ

S

δ
.

Thus 99% VaR is approximately 2.33

σ

L

.

Delta-gamma approximation

For nonlinear portfolios,

Δ
V
≈

∂
V

∂
t

Δ
t

+
δ

⊤

Δ
S
+

1
2

Δ
S

⊤

Δ
S

T
Δ
S
,

δ

i

=

∂
V

∂

S

i

,
    

Γ

i
j

=

∂
2
V

∂

S

i

∂

S

j

.

Let

Δ
S
=
C
Z
,
Z
∼
N
(
0
,
I
)
,
C

C

⊤

=

Σ

S

.

 Then

L
≈
a
−
(

C

⊤

δ
)

⊤

Z
+

1
2

Z

⊤

(

C

⊤

Γ
C
)
Z
,
    
a
=
−
(
Δ
t
)

∂
V

∂
t

.

Choose

C

 to diagonalize the quadratic term. If

−

1
2

C

⊤

Γ
C
=
U
Λ
U

⊤

,
C
=
C
U
,

Λ
=
diag
⁡
(

λ

1

,
.
.
.
,

λ

m

)
,

 and

b
=
−

C

⊤

δ
,

 then

Q
=
a
+
b

⊤

Z
+

Z

⊤

Z

Λ
Z
=
a
+

∑

j
=
1

m

(

b

j

Z

j

+

λ

j

Z

j

2

)
.

Use

P
(
L
>
x
)
≈
P
(
Q
>
x
)
.

Distribution of Q

The cumulant generating function

ψ
(
θ
)
=
log
E
[

e

θ
Q

]

 is

ψ
(
θ
)
=
a
θ
+

1
2

∑

m

j
=
1

(

θ

2

j

2

−

b

j

2

1
−
2
θ

λ

j

}

.

valid when

max
⁡

j

θ

λ

j

<
1

/

2
.

 The characteristic function is

ϕ
(
u
)
=

e

ψ
(
i
u
)

.

 Trans- form inversion gives

P
(
Q
≤
x
)
−
P
(
Q
≤
x
−
y
)
=

1
2

∫

0

y

∞

e

i
u
x

(

1
−
i
u
y

ω
−
1

ω
−
1

}

)
d
u
.

Choose

y

 large so

P
(
Q
≤
x
−
y
)

 is negligible.

Ptain MC and quantile

Ordinary MC generates market moves, revalues the portfolio, and estimates

P
(
L
>
x
)
≈

1
N

∑

N

i
=
1

1

{

L

i

>
x
}
.

The bottleneck is portfolio revaluation. With empirical CDF

F

L,n

(
x
)
=

1
n

∑

n

k
=
1

1

{

L

k

≤
x
}
,

estimate VaR by

x

p

=

F

L,n

−
1

(
1
−
p
)
.

8.2 Variance Reduction

Control variate

Evaluate both

L

i

 and

Q

i

 on each path. A control estimator for

P
(
L
>
x
)

 is

1
−

F

L,n

C
V

(
x
)
=

1
n

∑

N

i
=
1

1

{

L

i

>
x
}
−

β

(

1
n

∑

N

i
=
1

1

{

Q

i

>
y
}
−
P
(
Q
>
y
)
)
.

P
(
Q
>
y
)

 is known by transform inversion, and

y

 need not equal

x
.

 The meth- od can disappoint in the far tail:

L

i

 and

Q

i

 may be highly correlated, but their exceedance indicators may not be, and few exceedances make

β
 noisy.

Importance sampling

Large

Q

i

 occurs when

b

j

Z

j

 is large or when

λ

j

Z

j

2

 is large with

λ

j

>
0
.

 Define the twisted measure

d

P

θ

d

P

=

e

θ
Q
−
ψ
(
θ
)

.

Then

P
(
L
>
x
)
=
E
θ
[

e

−
θ
Q
+
ψ
(
θ
)

1

{

L
>
x
}

]
.

Under

P

θ

,

Z

 remains normal with diagonal covariance:

Z
∼
N
(
μ
(
θ
)
,
Σ
(
θ
)
)
,
    
μ

j

(
θ
)
=

θ

j

b

j

1
−
2
λ

j

θ
,
    

σ

j

2

(
θ
)
=

1
1
−
2
λ

j

θ

.

For

θ
>
0
,

 the mean shifts in the sign of

b

j

 and variance increases when

λ

j

>
0
.